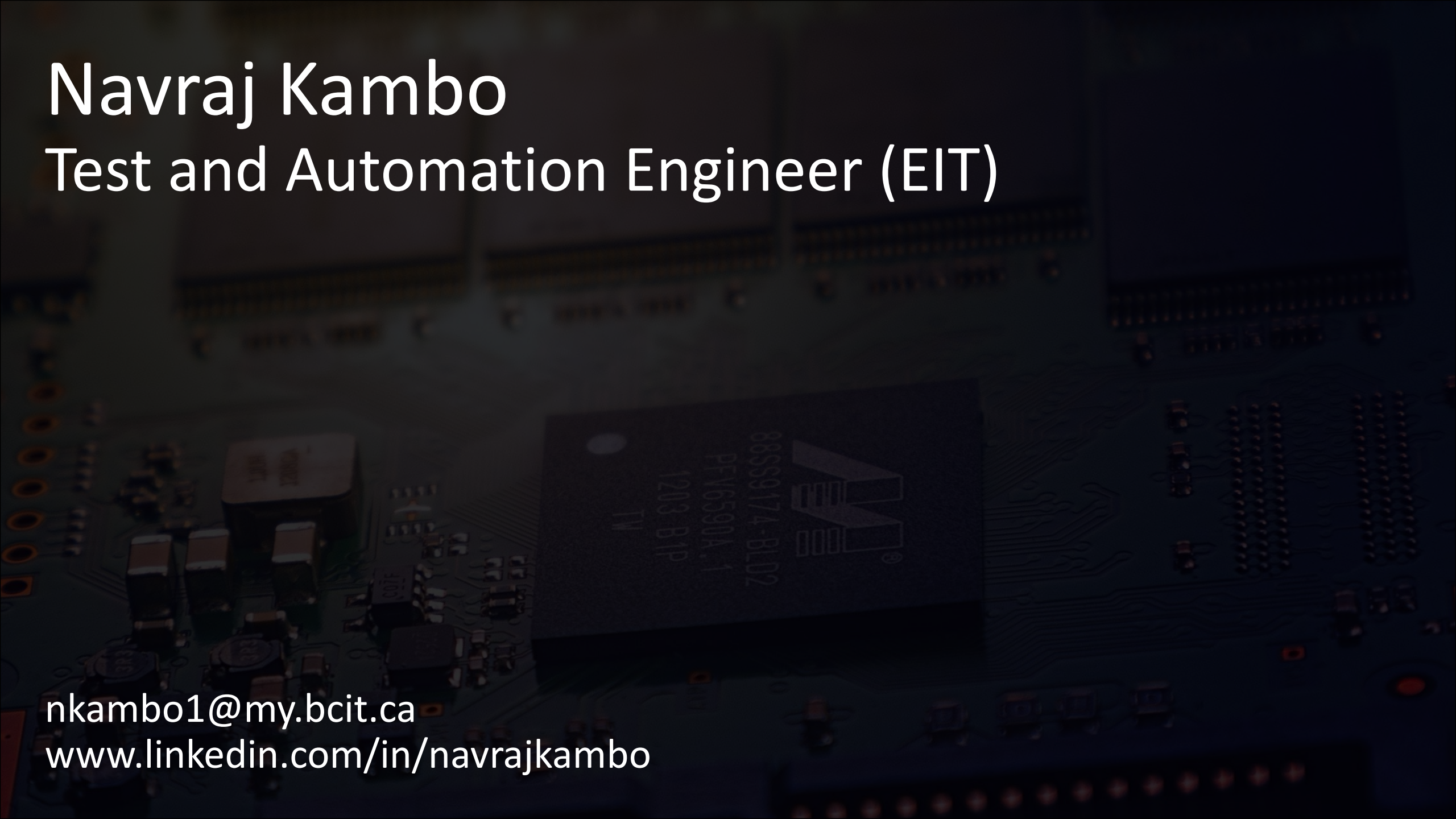




Navraj Kambo

Test and Automation Engineer (EIT)

nkambo1@my.bcit.ca

www.linkedin.com/in/navrajkambo

Navraj Kambo | Overview

- 3 About Me
- 4 Digital Speech Decoding (WTS Wireless Technical Services Inc. / FortisBC)
- 5 NDART (British Columbia Institute of Technology)
- 6 Autonomous Vineyard Harvesting (British Columbia Institute of Technology SoE)
- 7 Apex Motion Control
- 8 Packaging Line(Apex Motion Control)
- 9 DecoBot (Apex Motion Control)
- 10 Wellons Canada
- 11 Continuous Drying System (Wellons Canada)
- 12 Delta Intelligent Building Technologies (Canada) Inc.
- 13 RED 5 Edge (China) (DIBT)
- 14 Test & Automation (DIBT)



Navraj Kambo | About Me

I'm an electrical engineer who graduated in 2019 from BCIT. I have experience in robotics, automation, and electronics manufacturing. I enjoy collaborating in multi-disciplinary teams, and outside of work, I like to play sports and keep myself active. I'm currently working at DIBT Canada (Inc) and am working through my master's degree under UBC's ECE department.

Education



**BEng Electrical Engineering
Research Intern (4 months)**



Experience



4-month co-op



Design Engineer



Electrical Engineer EIT

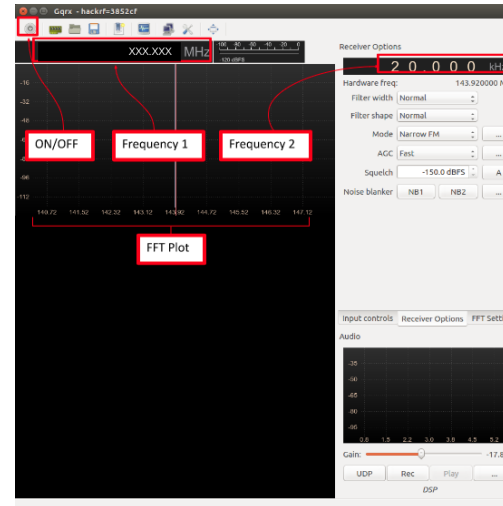


Test & Automation Engineer

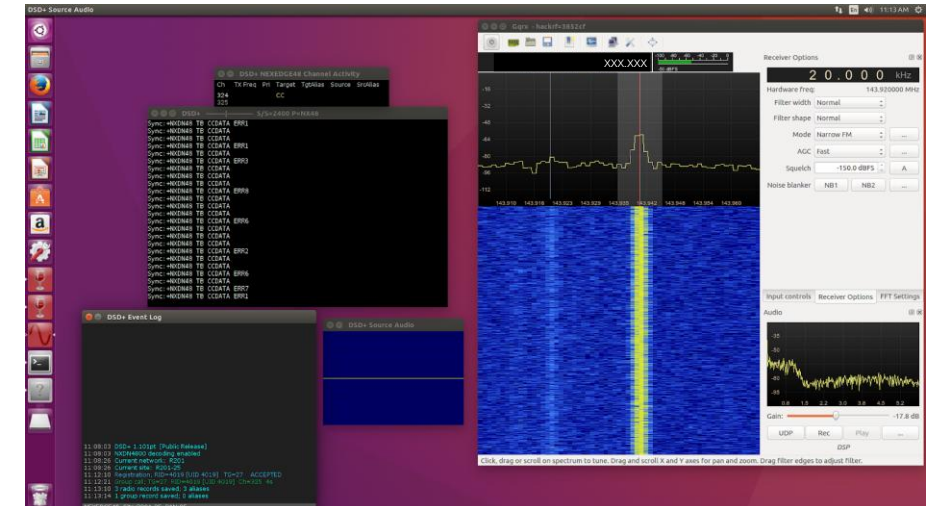
Navraj Kambo | Digital Speech Decoding

```
wts-nav@KRYPTONITE: ~  
wts-nav@KRYPTONITE:~$ hackrf_info  
Found HackRF board 0:  
USB descriptor string: 000000000000000909864c8383852cf  
Board ID Number: 2 (HackRF One)  
Firmware Version: 2015.07.2  
Part ID Number: 0xa000cb3c 0x00594f4f  
Serial Number: 0x00000000 0x00000000 0x909864c8 0x383852cf  
wts-nav@KRYPTONITE:~$
```

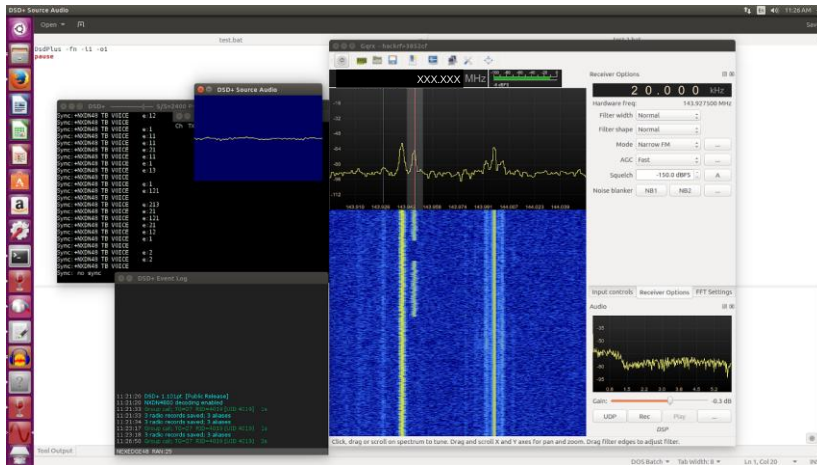
HackRF in Ubuntu



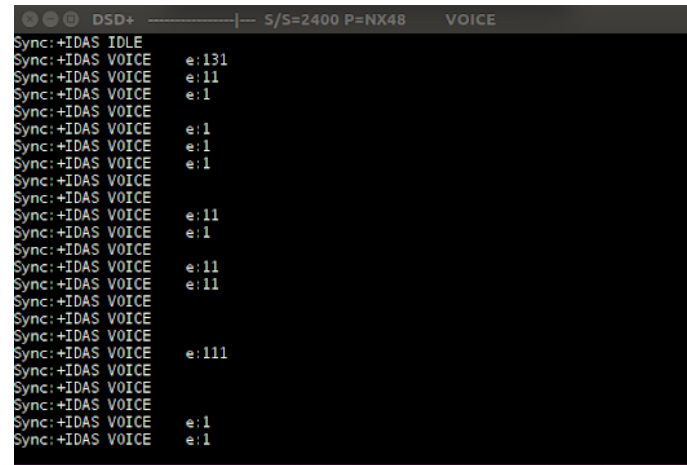
GQRX SDR



Decoding Control Channel (Carrier)



Decoding Traffic Channel (Side-band)

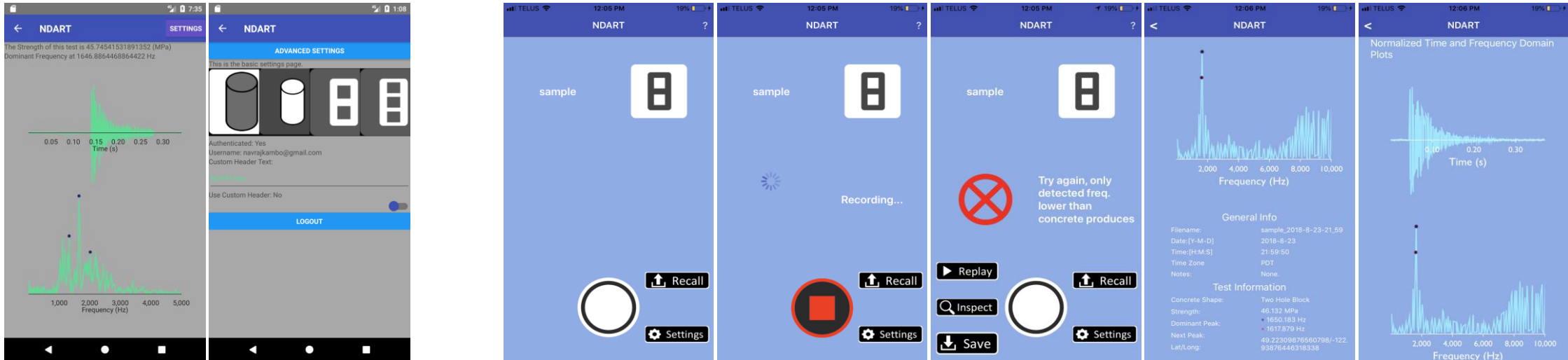


Sync Window on Traffic Channel



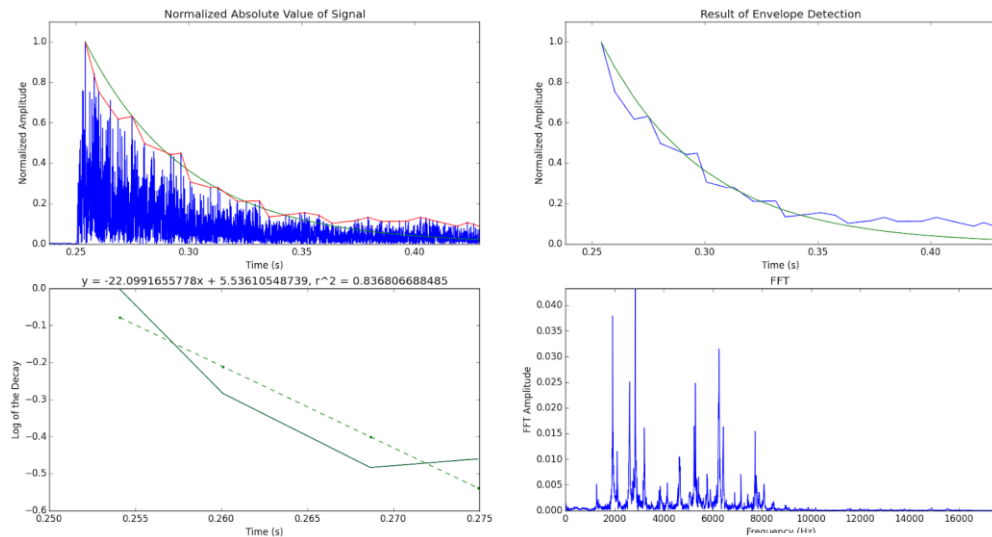
HackRF One

Navraj Kambo | NDART



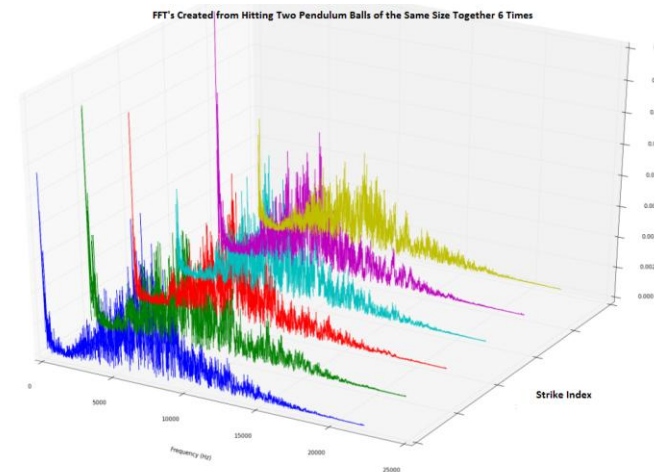
NDART app on Android

NDART app on iOS

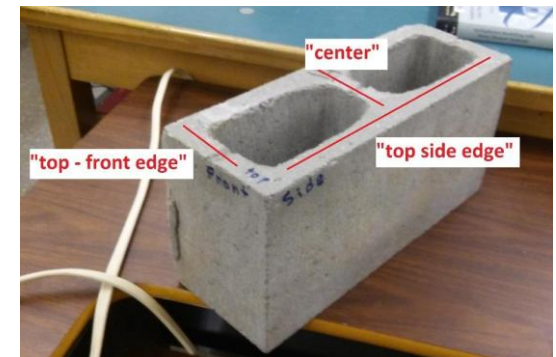


Strike Decay Plot

$$N(t) = N_0 e^{-\lambda t}$$



FFTs at Different Locations

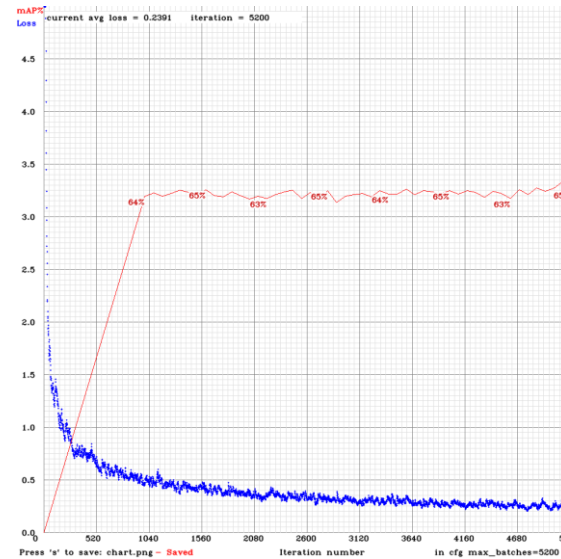


Concrete Block

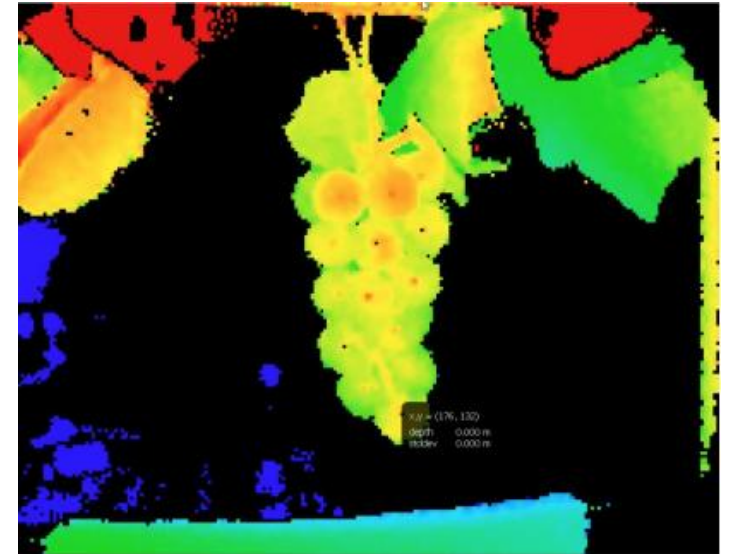
Navraj Kambo | Autonomous Vineyard Harvesting



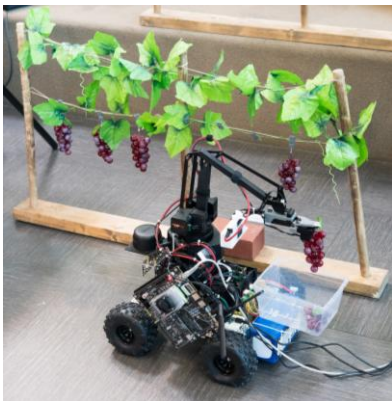
Project Expo



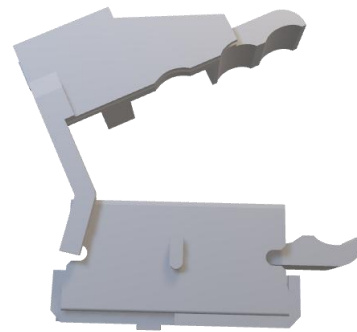
ML Dataset Training



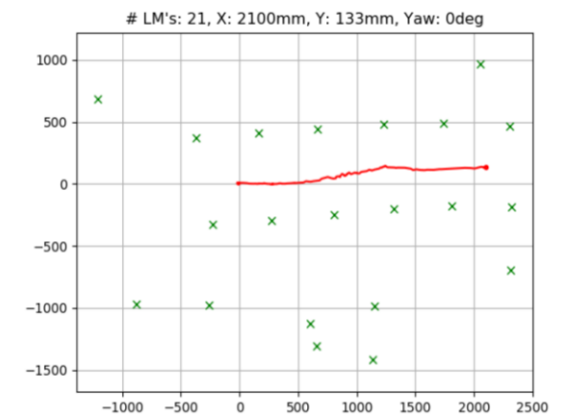
Grape Cluster TOF Image



Vinebot

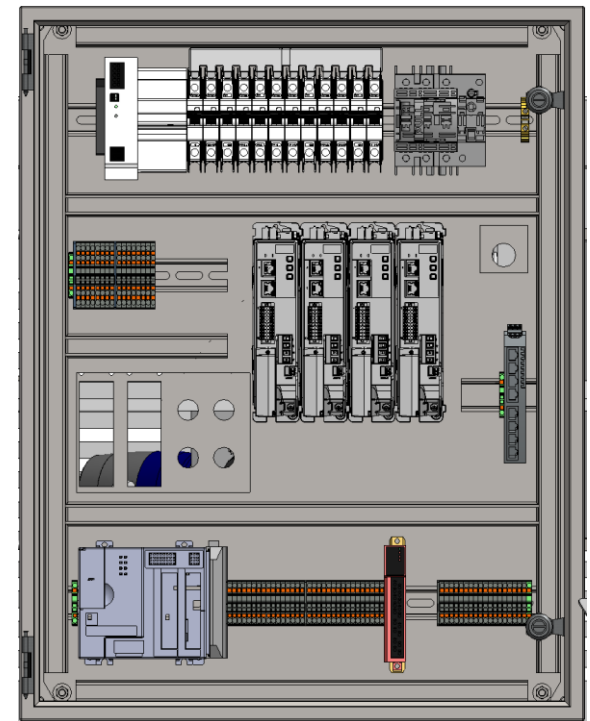
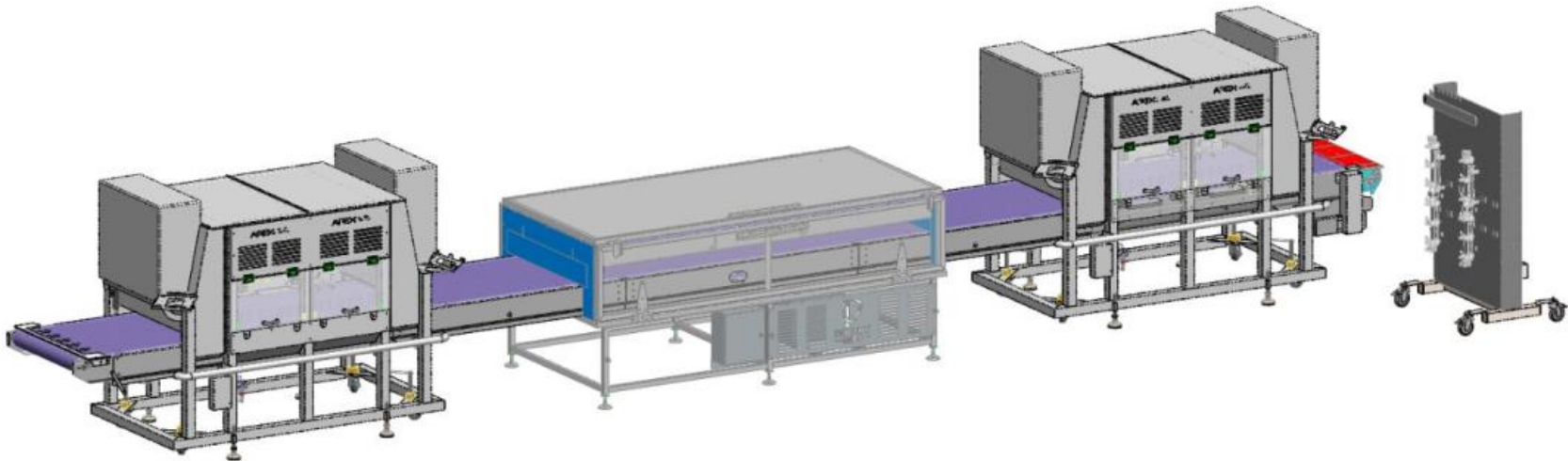


Vinebot Stem Cutter

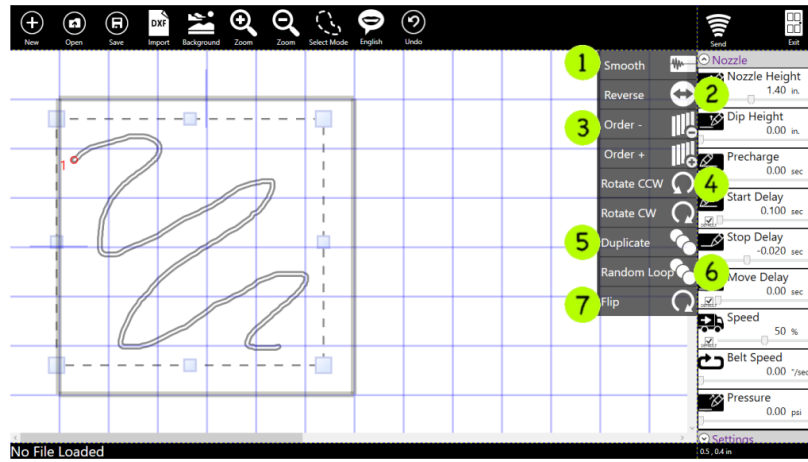


SLAM Path Planning

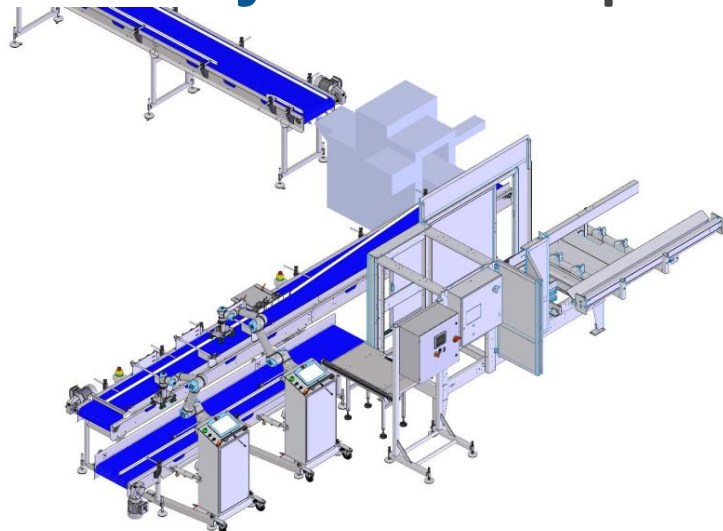
Navraj Kambo | Apex Motion Control - Design Engineer



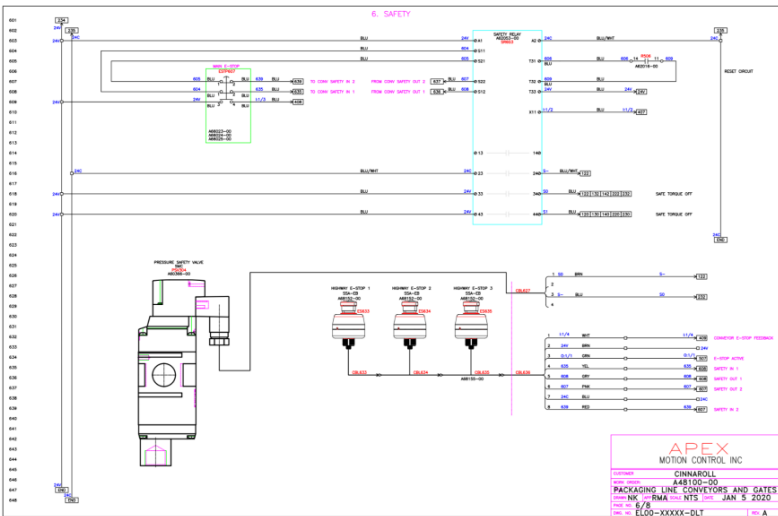
Robotics and Automation (Food and Beverage Industry)



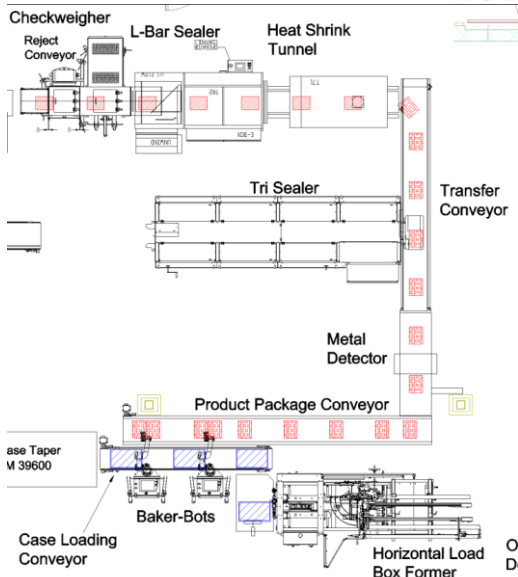
Navraj Kambo | Packaging Line



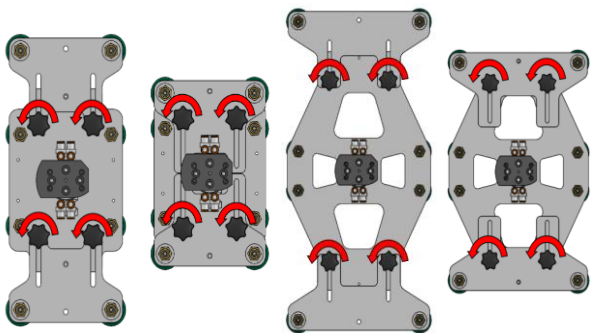
SolidWorks Assembly



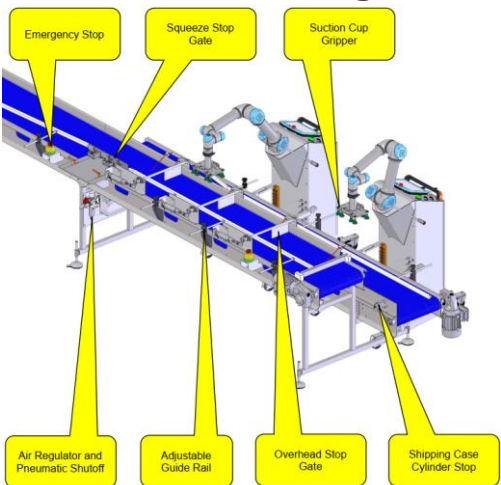
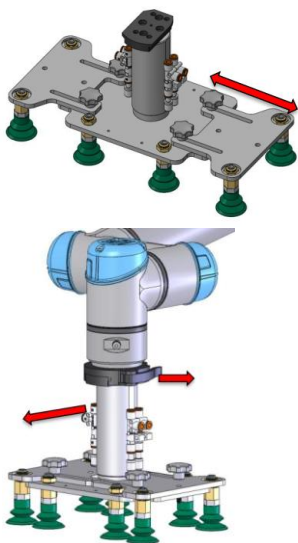
Electrical Design



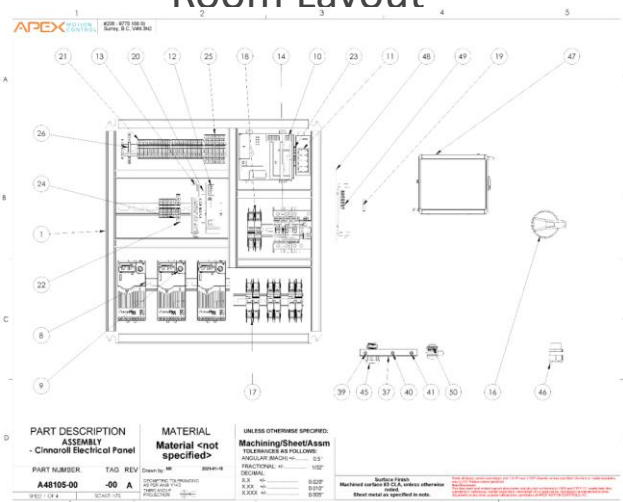
Room Layout



Adjustable Tooling

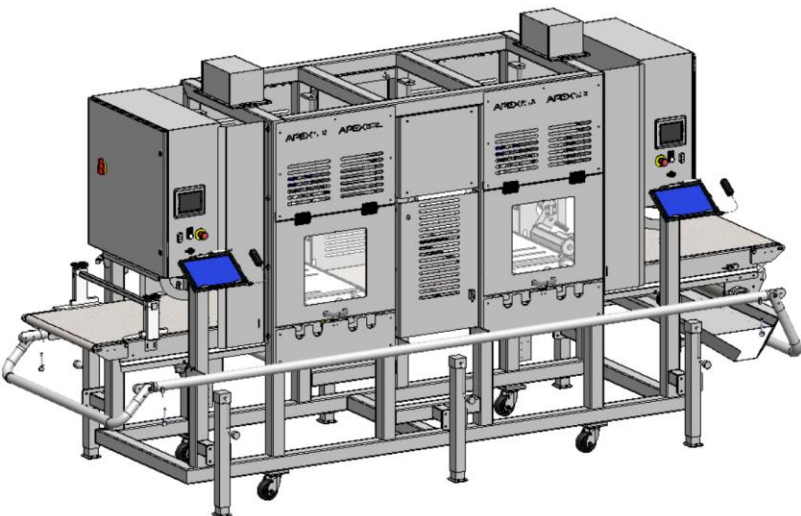


Pick & Place Cell

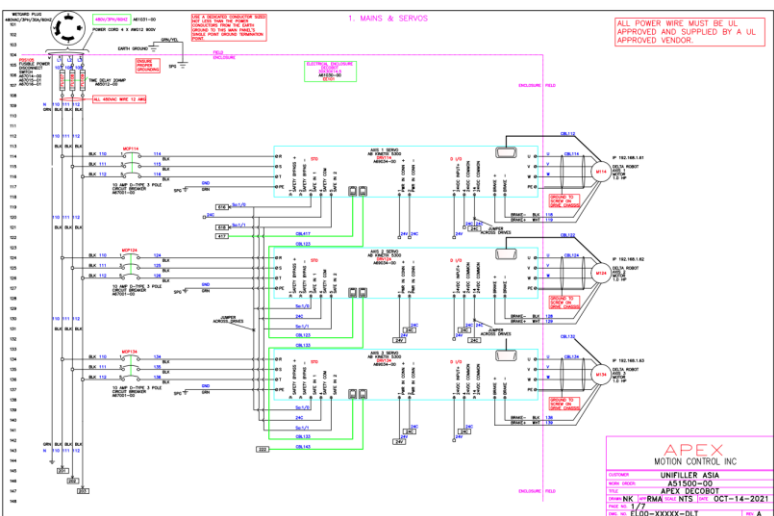


Electrical Panel Layout

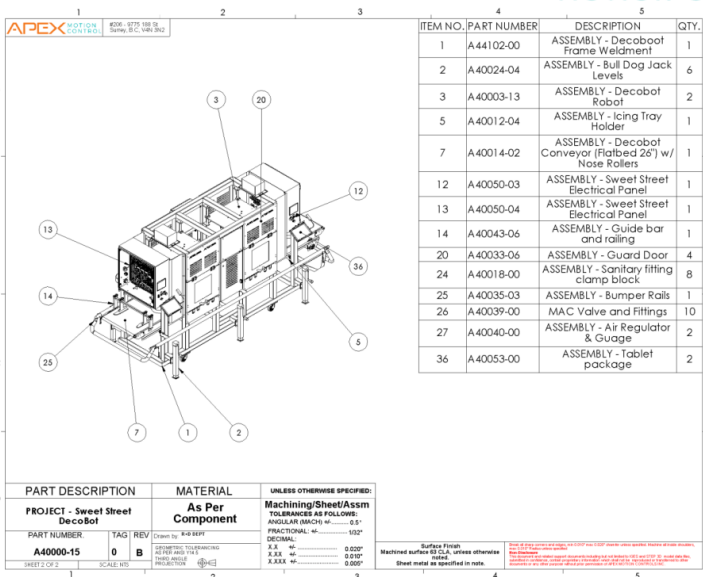
Navraj Kambo | DecoBot



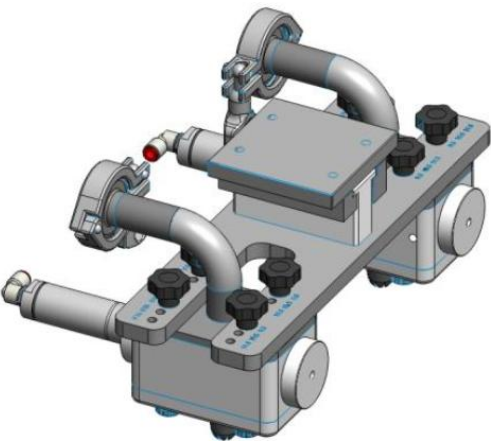
SolidWorks Assembly (Dual DecoBot)



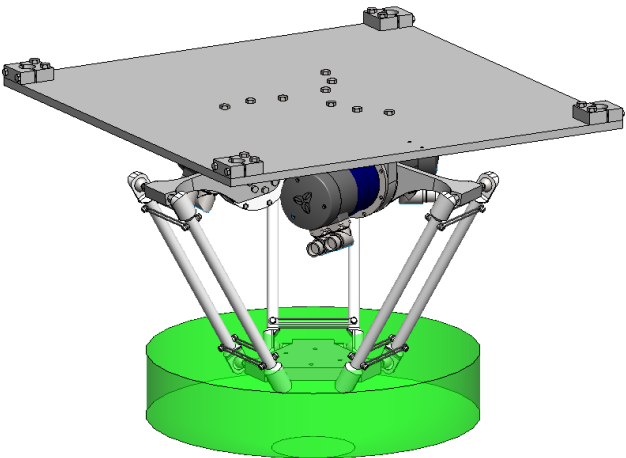
Electrical Design



Machine Drawing



Robot Decorating Nozzle

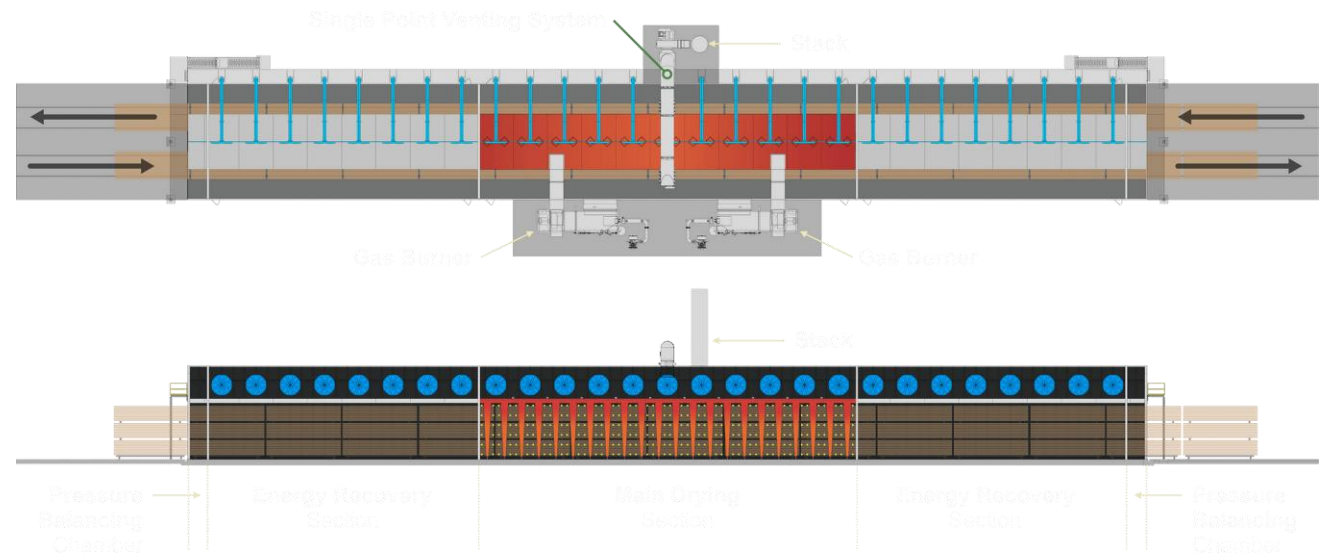


Delta Robot

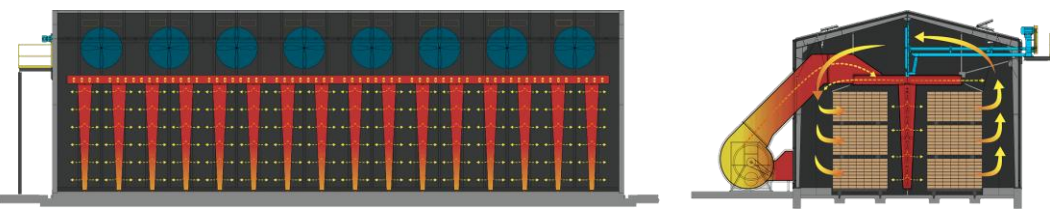


Actual Robot

Navraj Kambo | Wellons Canada – Electrical Engineer



Wood Drying Kilns and Energy Systems



FuelSystem

Fuel System

Bucket Elevator

Reclaimer Drag Chain

0" Depth

Vibrating Conveyor

Fuel System Auto / Stop

AUTO STOP

Silo Antitator

Silo Feedout Auger

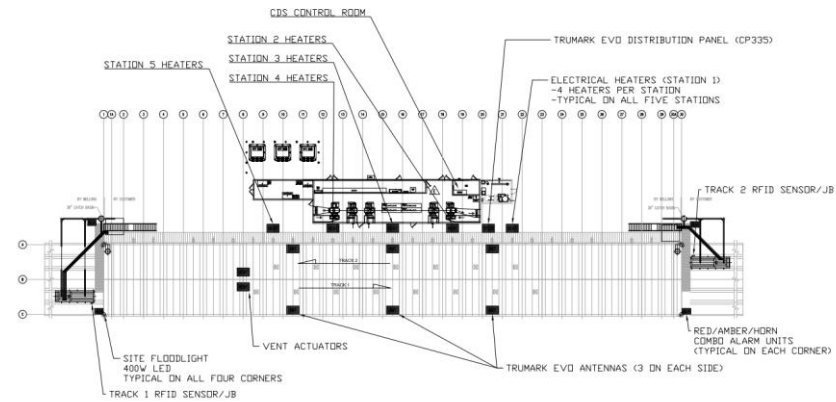
Fuel Transfer Conveyor

To Fuel Metering Bin

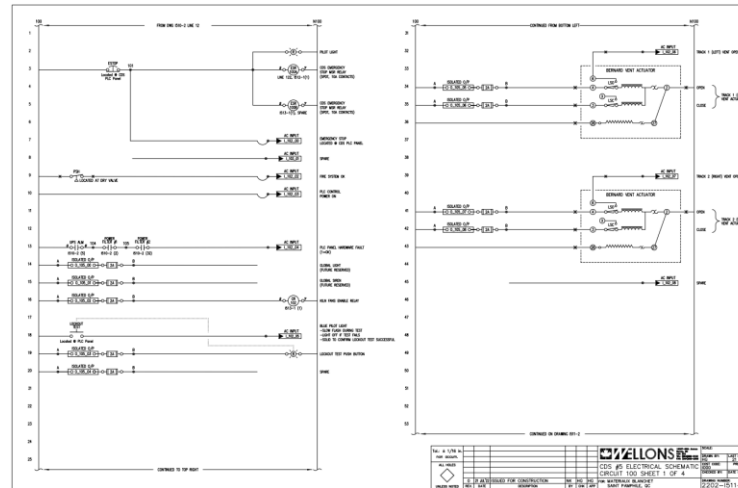
Fuel System Timers			
Reclaimer Speeds			
SC0254 Very Low Speed	12 Hz	SC0254 Low Speed	12 Hz
SC0254 Medium Speed	12 Hz	SC0254 High Speed	12 Hz
Reclaimer Unloader Stop Times			
MX0251 Stop Delay	0 Sec	MX0255 Stop Delay	0 Sec
SC0254 Stop Delay	0 Sec		
Feed Out Auger Start Delay		Reclaimer High Alarm Delay	
MX1416 Start Delay	3 Sec	LSH0263 Delay Time	0 Sec
Silo Feedout Bin High Level		LSH1351 Call for Fuel	
LSH1418 Alarm Time	10 Sec	LSH1351 Off Delay	0 Sec
Reclaimer Lube SV			
EV0269 On Time	0 Sec	EV0269 Off Time	0 Sec
Silo Ratchet Drive			
EV1415A Ext Time	10 Sec	EV1415A Ret Time	10 Sec
EV1415B Ext Time	10 Sec	EV1415B Ret Time	10 Sec



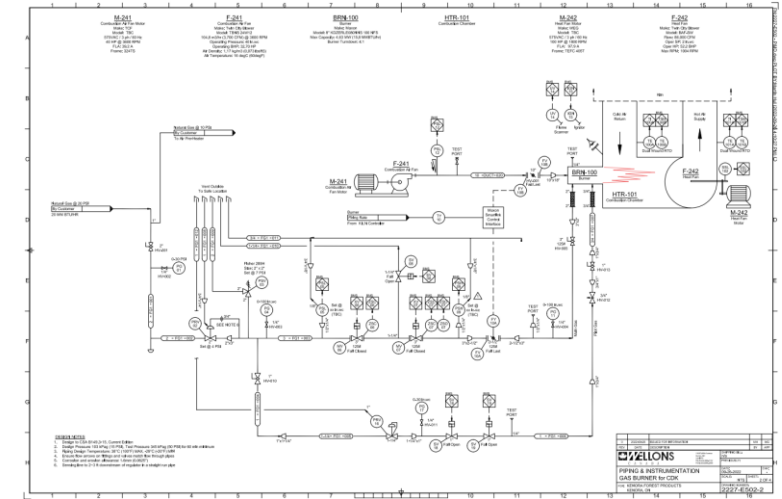
Navraj Kambo | Continuous Drying System



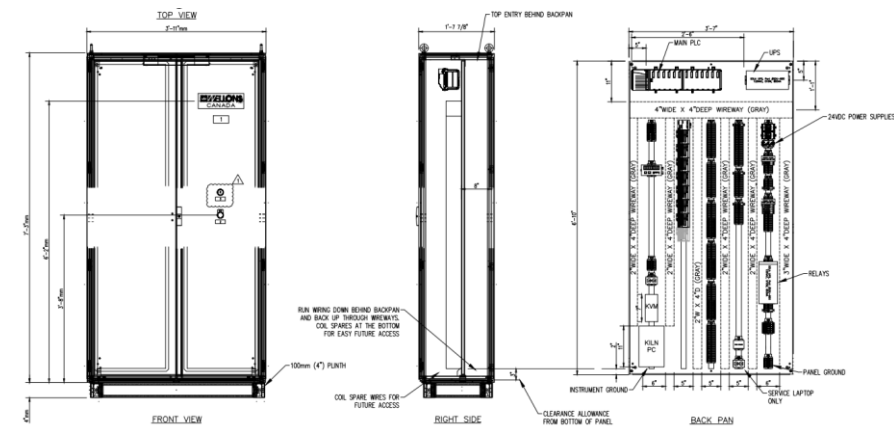
Kiln Layout Drawings



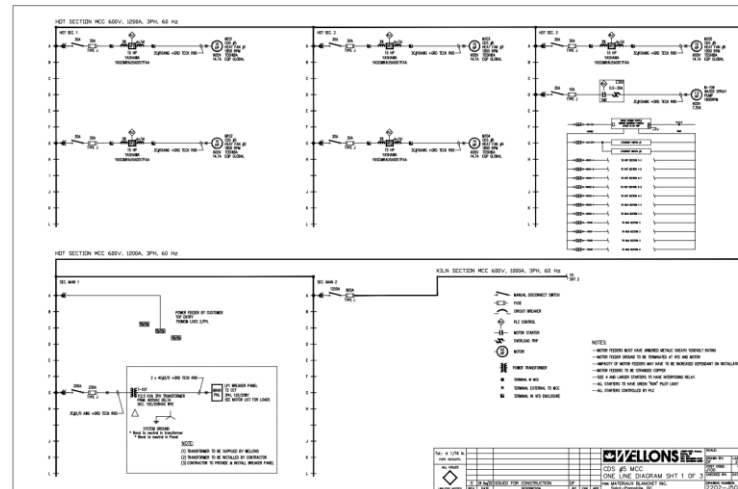
Control Schematics



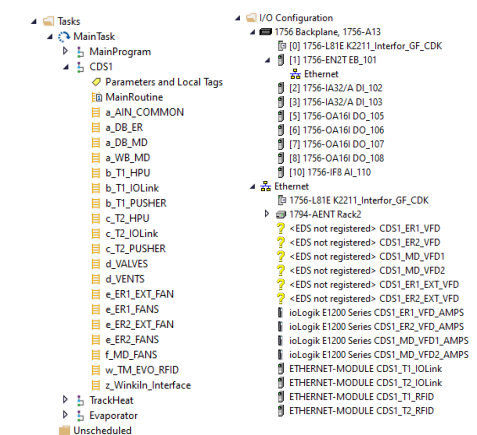
Gas Train P&IDs



Control Panel Design

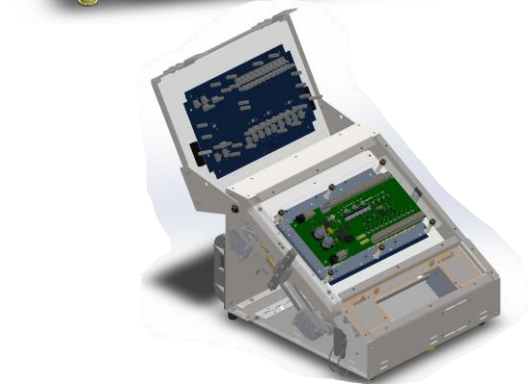
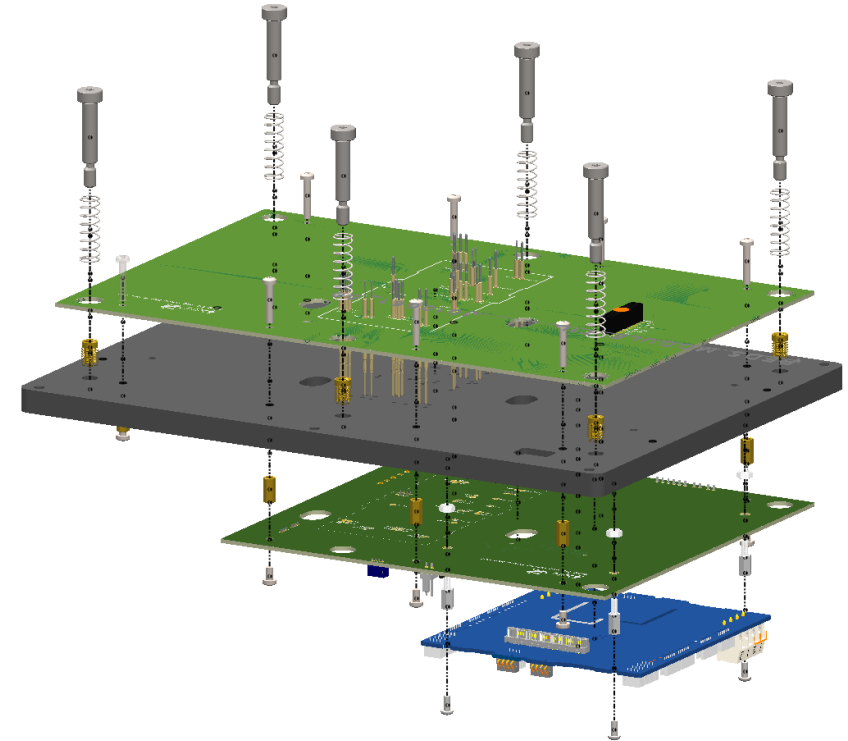
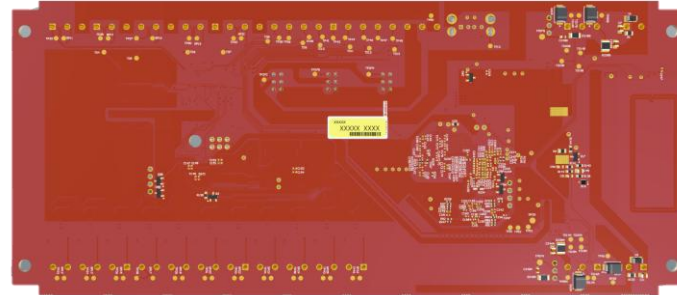
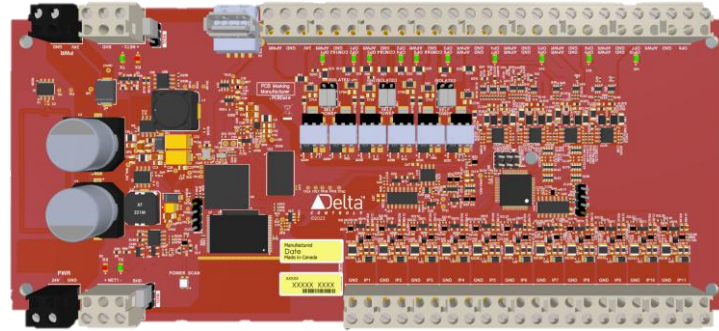
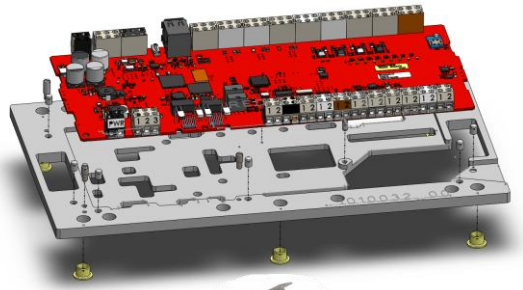


MCC Layout

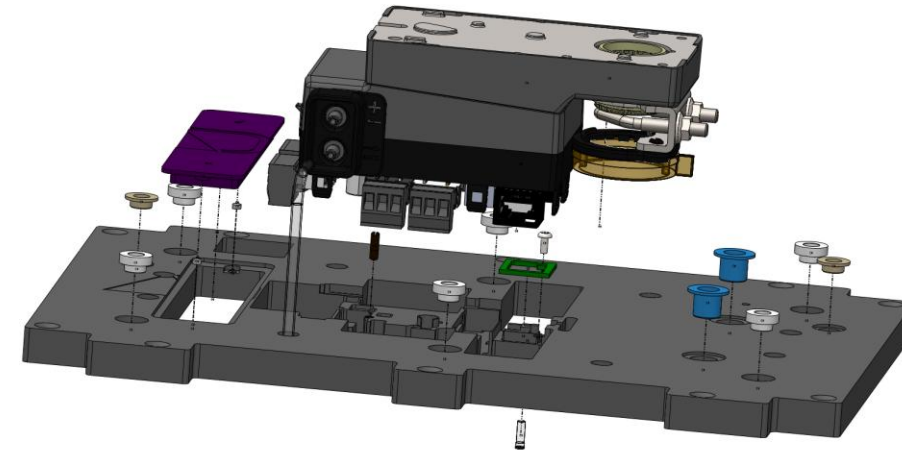
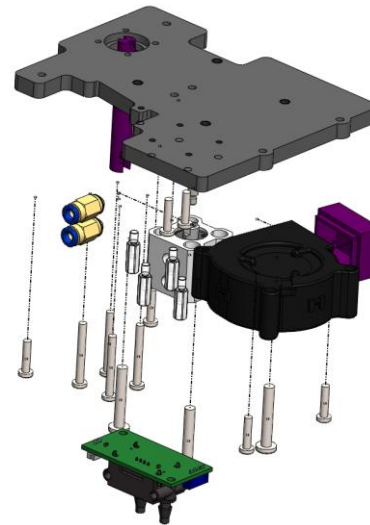
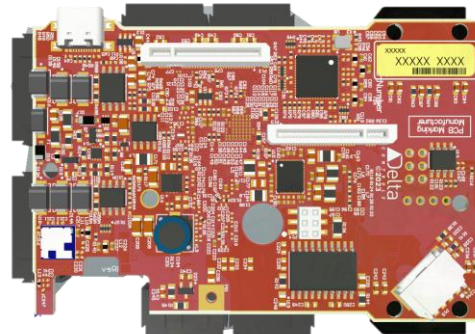
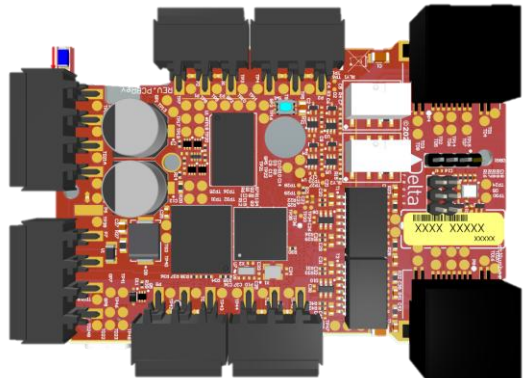


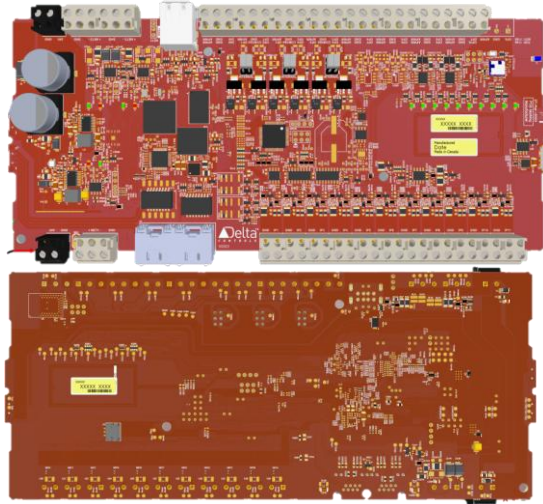
PLC Programming

Navraj Kambo | DIBT – TAG Engineer

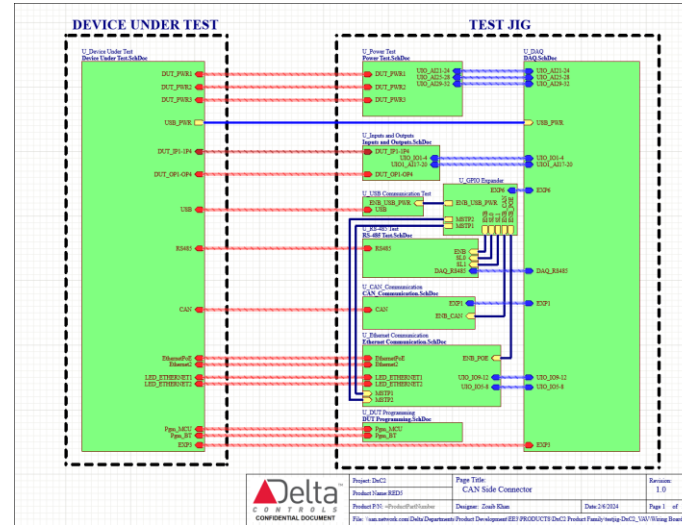


HVAC Controllers and Smart Thermostats (Testing Group)

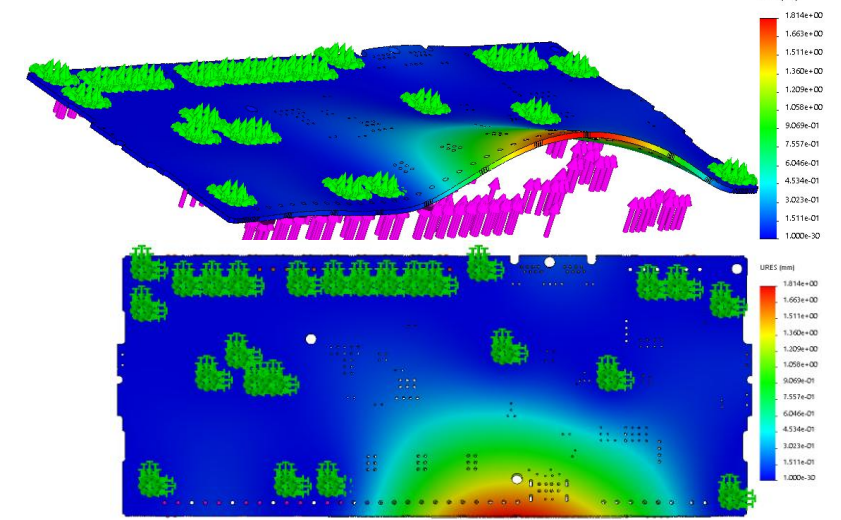




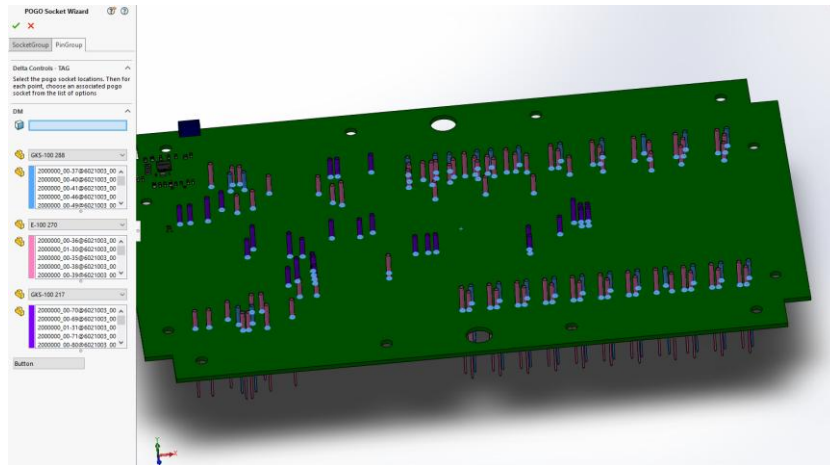
Hardware Review



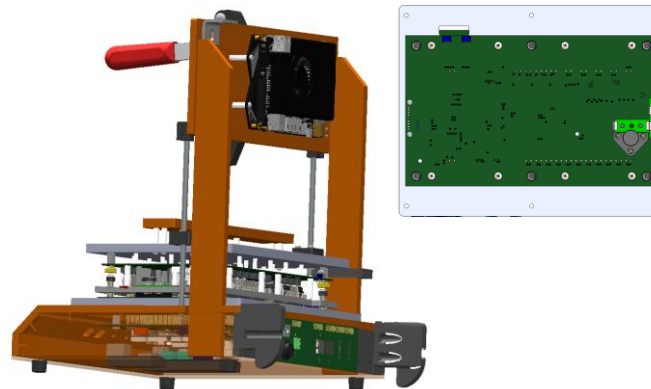
Test Jig Schematic Review



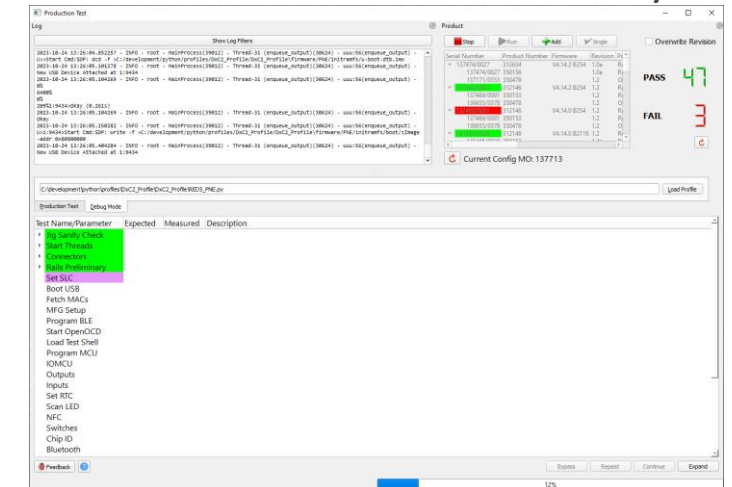
POGO Pin Static Analysis



POGO Pin/Socket Selection

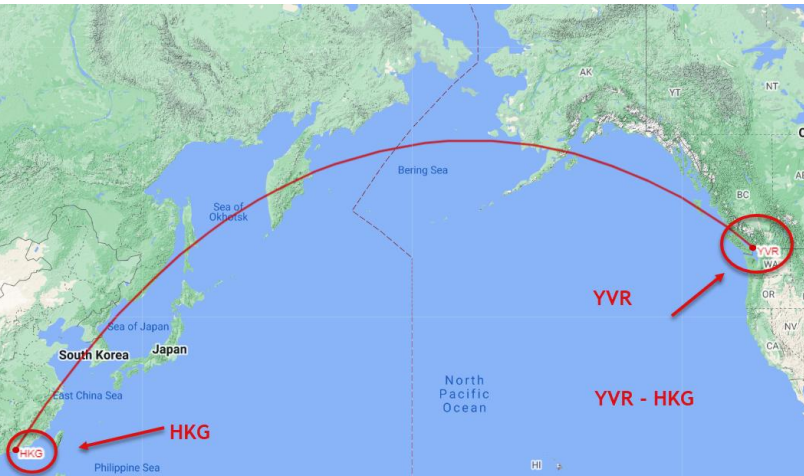


Mechanical Design

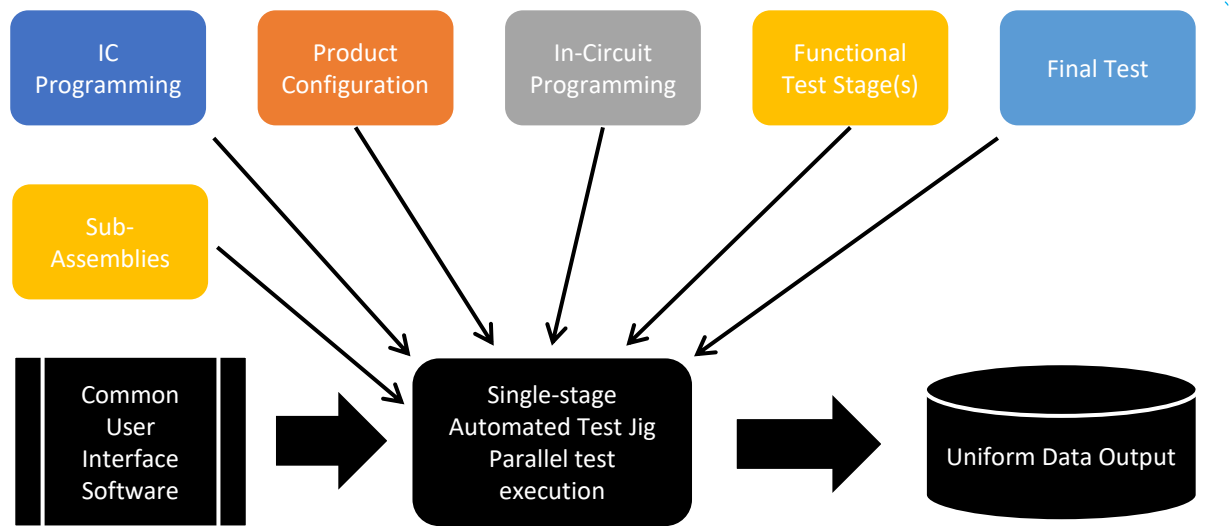
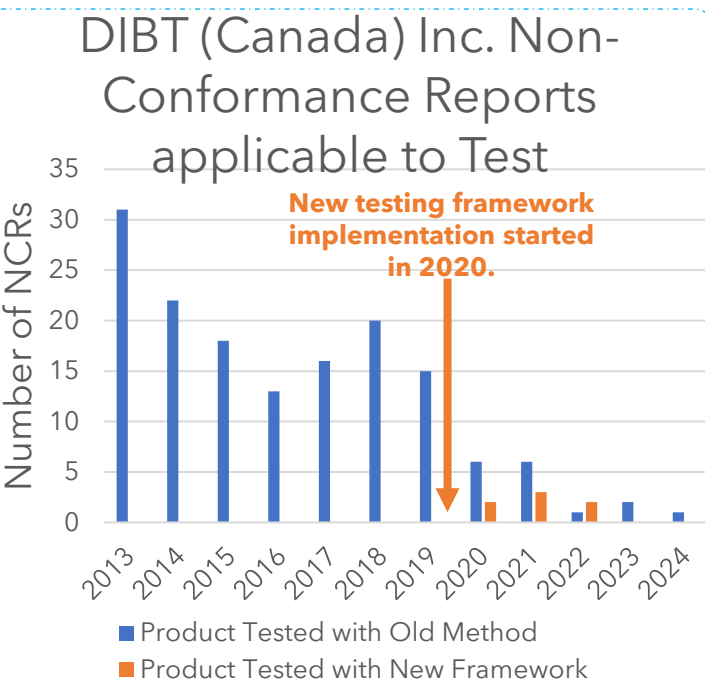
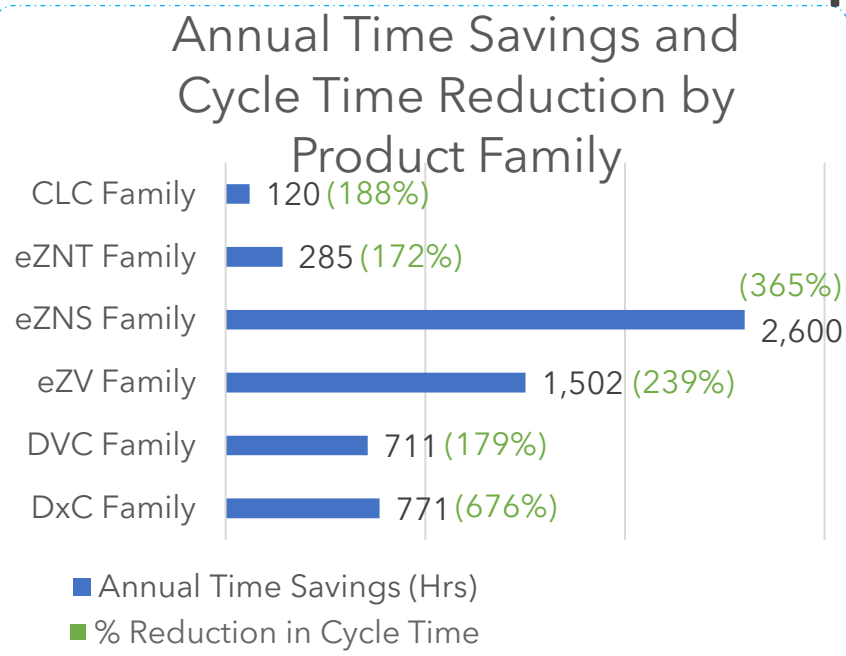
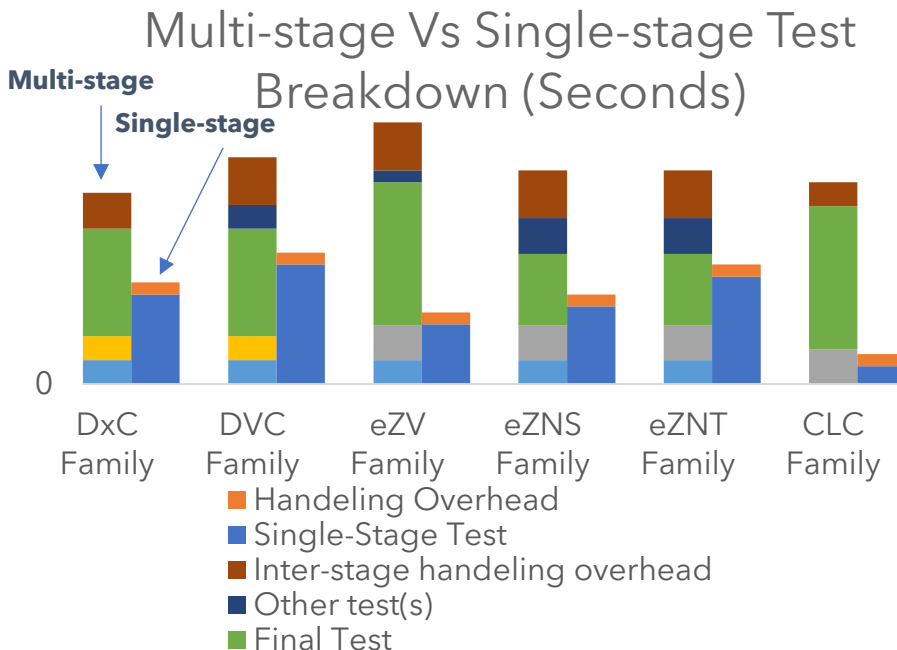


Test Firmware and Framework(UI)

Navraj Kambo | RED 5 Edge (China)



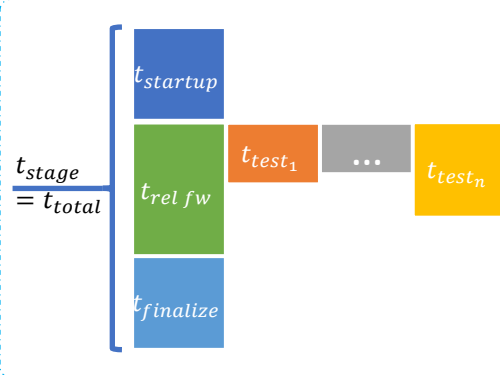
Navraj Kambo | Test and Automation Group

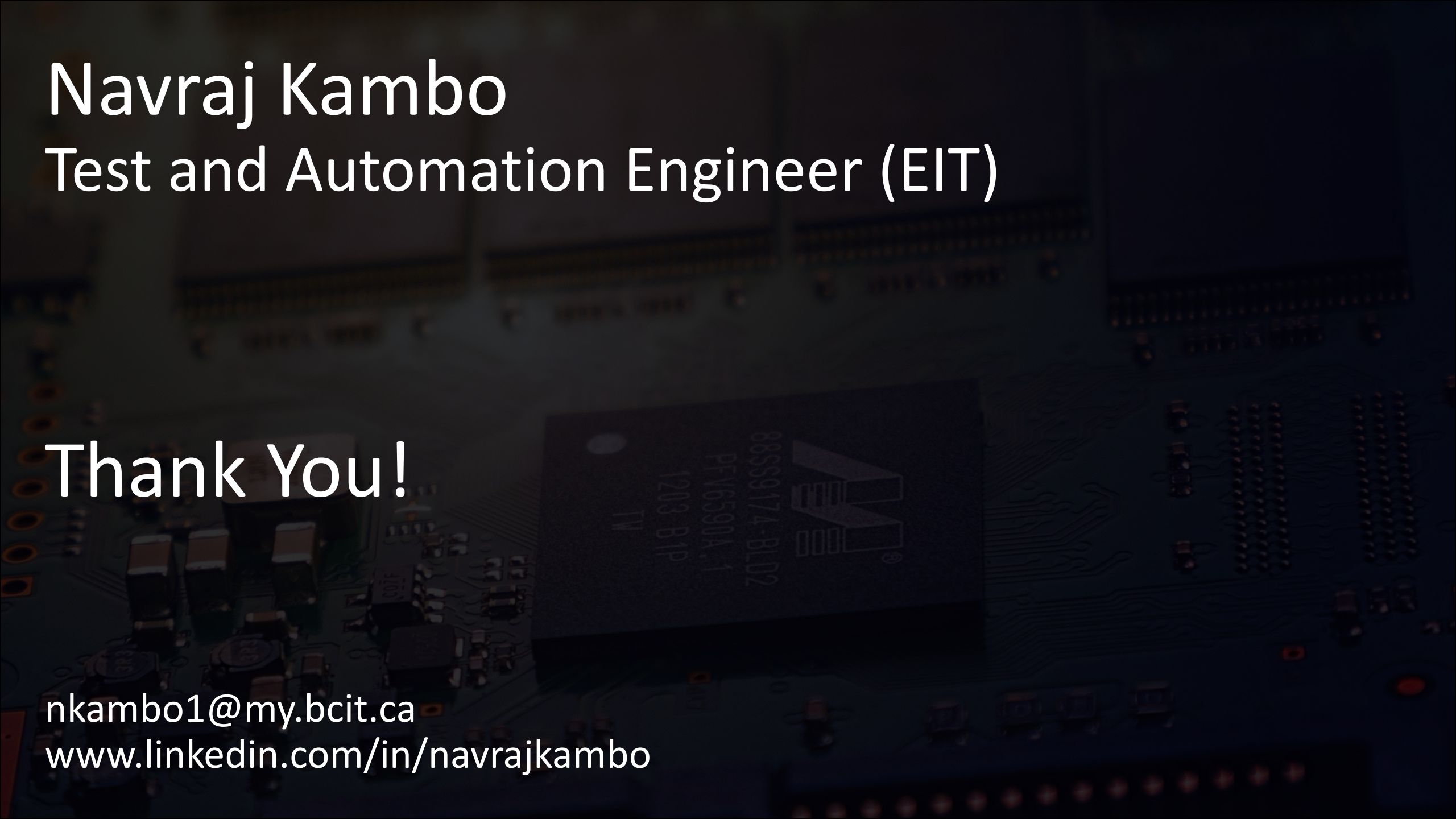


The DIBT (Canada) Inc. Test and Automation team developed a flexible and efficient framework (software, firmware, hardware, and data analysis) that integrates multiple production stages into a single streamlined process.

Key benefits:

- Saves 6,000+ hours annually
- Drastically improves test coverage and data visibility
- Reduces human intervention and standardizes workflows
- Eliminates the need for specialized product training.





Navraj Kambo

Test and Automation Engineer (EIT)

Thank You!

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